

Constraint Based Problem

OUTPUT

```
Output
===== AI Assessment =====
1. Hospital Doctor Scheduling
2. Robot Navigation using BFS
3. Autonomous Rescue Robot
4. Fraud Detection System
5. Convex Hull using Brute Force
6. Exit

Enter your choice : 1

Doctor Scheduling using Backtracking

Valid Schedule
D1 -> Morning
D2 -> Afternoon
D3 -> Night

===== AI Assessment =====
1. Hospital Doctor Scheduling
```

```
Output Clear
===== AI Assessment =====
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Enter your choice : 2

Robot Navigation using BFS

Shortest Path
[(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 3), (2, 4), (3, 4), (4, 4)]
Total Cost = 8

===== AI Assessment =====
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```

```
Output Clear
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Enter your choice : 3

Autonomous Rescue Robot using Uniform Cost Search

Goal Reached
Path = [(0, 0), (1, 0), (2, 0), (2, 1), (3, 1), (4, 1), (4, 2), (4, 3),
        (4, 4)]
Minimum Cost = 8

===== AI Assessment =====
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```

```
Output Clear
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Enter your choice : 4

Fraud Detection System

Enter Transaction Amount: 295000
Enter Location (Local/Foreign): local
Enter Number of Transactions Today: 7

Transaction Status : Fraudulent

===== AI Assessment =====
```

Output

===== AI Assessment =====

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Enter your choice : 5

Brute Force Convex Hull

Convex Hull Edges

((1, 1), (2, 5))

((1, 1), (6, 1))

((2, 5), (4, 6))

((4, 6), (6, 1))

Output

Convex Hull Edges

((1, 1), (2, 5))

((1, 1), (6, 1))

((2, 5), (4, 6))

((4, 6), (6, 1))

===== AI Assessment =====

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Enter your choice : 6

Program Ended

=== Code Execution Successful ===